



PM04

SDT621

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Completed Declaration of Authenticity

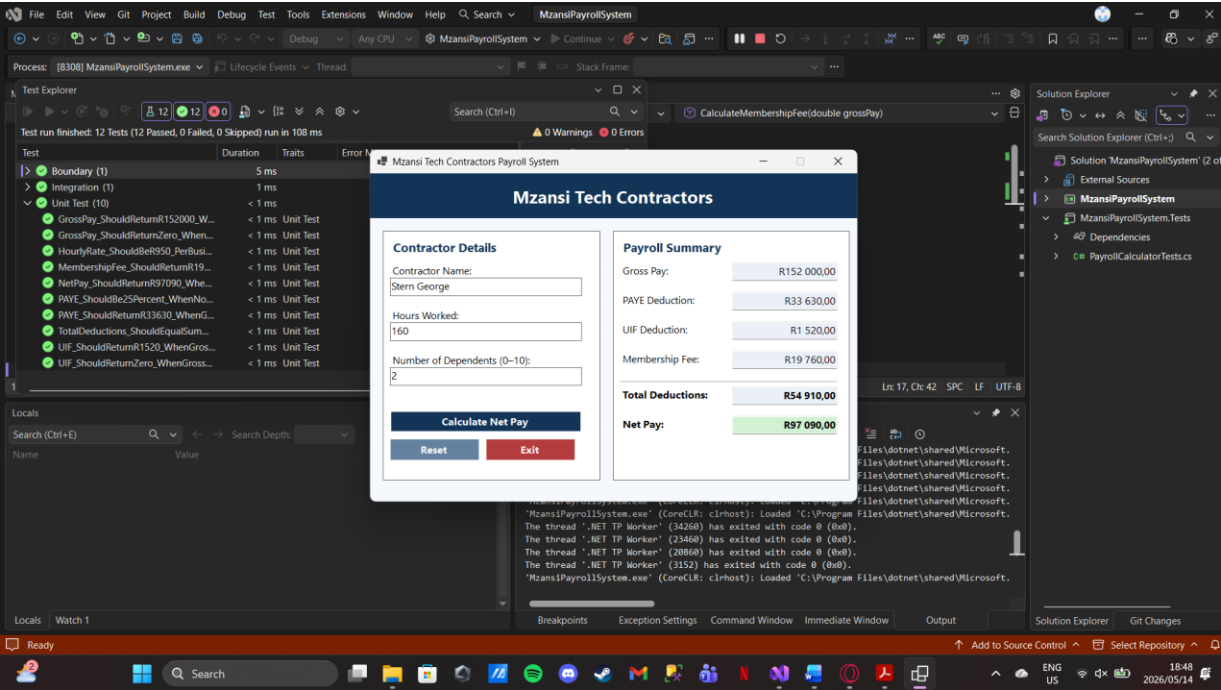
I, **Stern George** hereby
declare that the contents of this assignment **PM04** is entirely my own work except for the following documents:
(List the documents and page numbers of work in this portfolio that were generated in a group)

Activity	Date

Signature:  _____

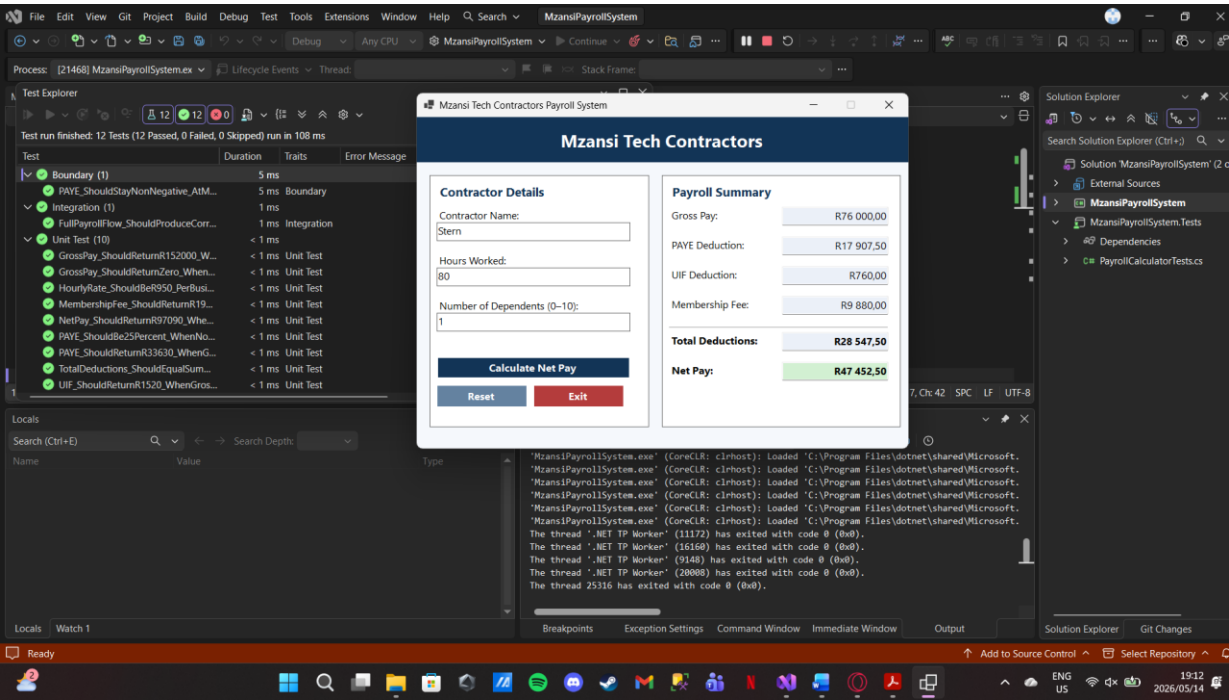
1. Unit Testing

Test Case	Input	Expected Output	Result
Gross Pay	Hours 160, Rate R950	R152 000	Pass
UIF	Gross R152 000	R1 520 (1%)	Pass
Membership	Gross R152 000	R19 760 (13%)	Pass
Net Pay	All above	R97 090	Pass



2.) Integration Testing

Integration testing was performed to verify that the three components of the payroll system work correctly together. The user inputs fields, the payrollcalculator class and the output display fields. Each test confirmed that data flows correctly from one component to the next without loss or corruption.



3.) System Testing

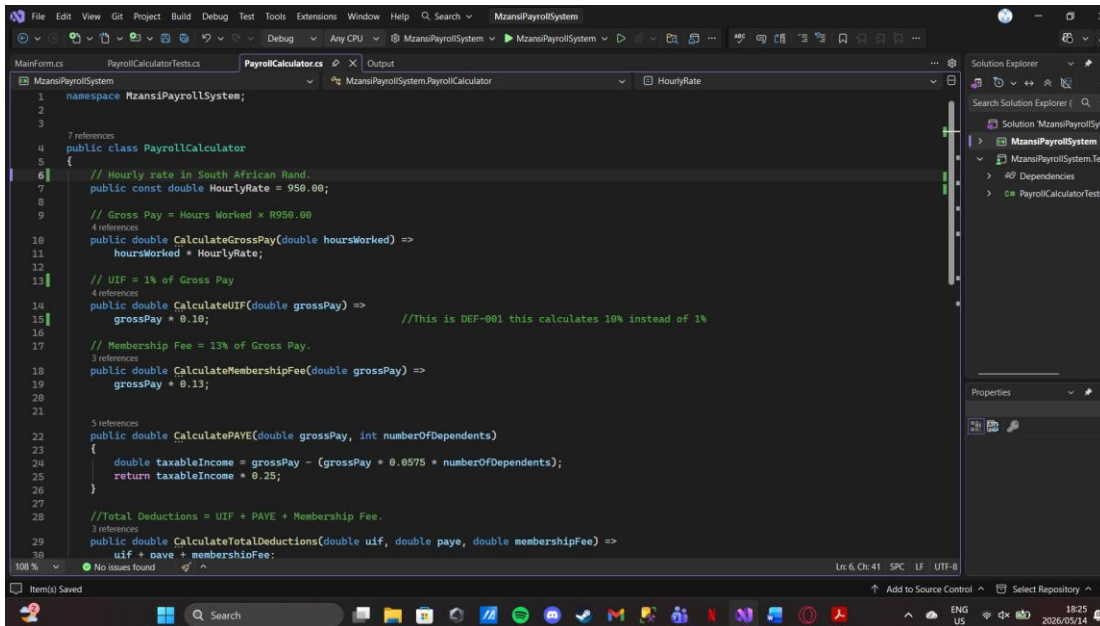
Step	Action	Expected Result
1	Launch the application	The Mzansi tech contractor payroll system window opens without error, all input fields are empty and all output fields are blank
2	Enter contractor name "Stern George", hours worked "160", and dependents "2", then click calculate Net Pay	All of the output fields populate correctly :Gross pay = R152 000, UIF = R1 520, Membership fee = R19 760, R33 630, Total deductions = R54 910, Pay = R97 090.
3	Clear all fields by clicking the reset button	All input and output fields return to empty / blank cursor moves to the contractor name field.
4	Leave all field empty and click calculate.	A validation error message appears: "Contractor name cannot be empty" – no calculations are performed.
5	Enter "Enter" in the Hours worked field and "-3" in the number of dependents field, then click calculate net pay	A validation error message appears: "Number of dependents must be a whole number between 0 and 10" – no calculations are performed.

4.) Retesting

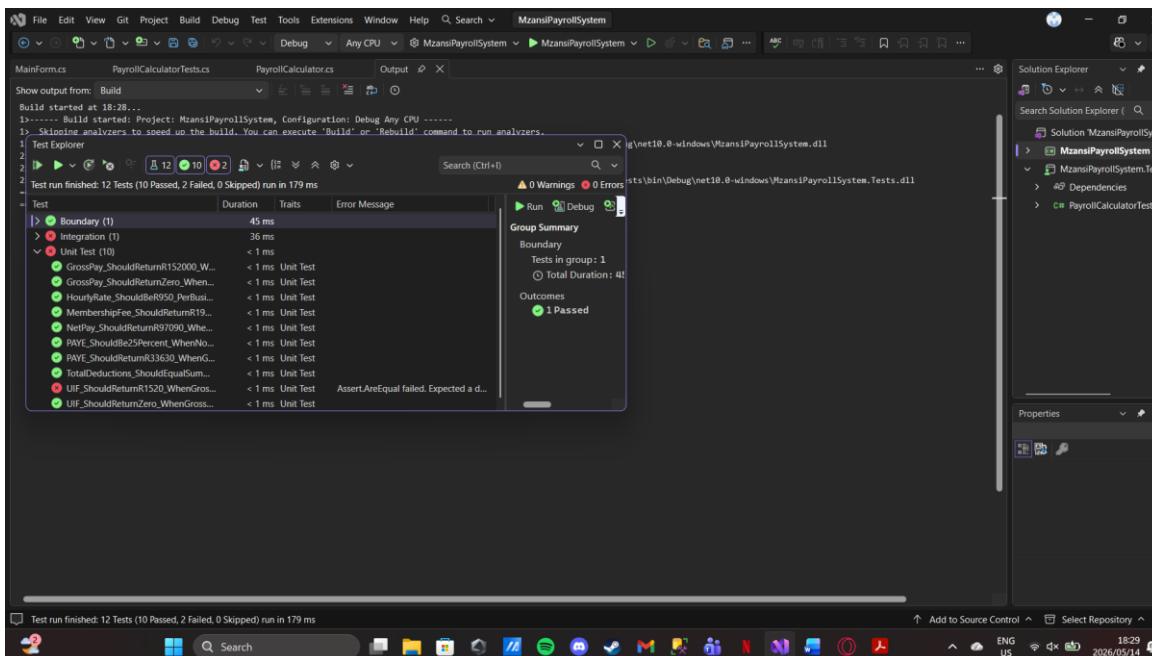
Defects:

Field	
Defect ID	DEF-001
Description	UIF deduction calculates 10% instead of the required 1%
Steps to reproduce	Enter 160 hours, 2 dependents, click calculate, UIF shows R15 200.00 instead of R1 520.00
Expected result	R1 520 (1% of R152 000)
Actual result	R15 200 (10% of 152 000)
Root cause	Multiplier in calculateUIF was 0.10 instead of 0.01

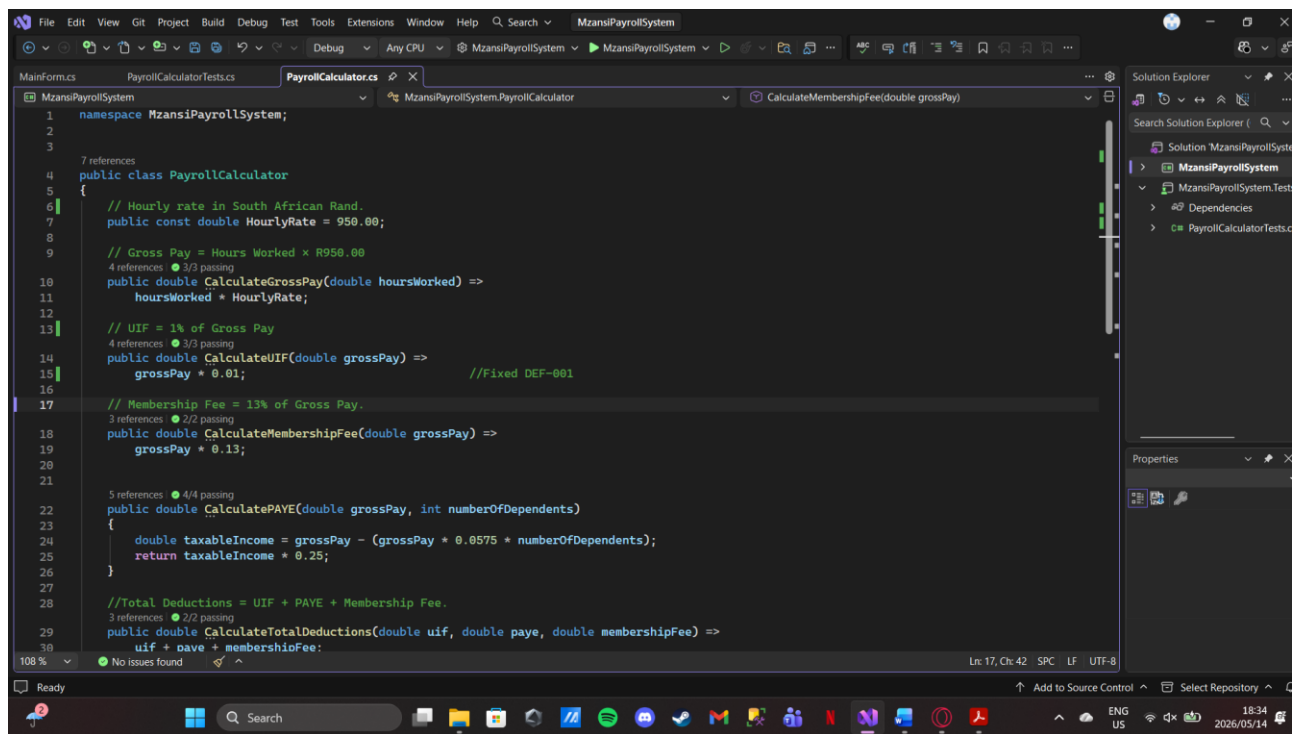
Screenshot of error in code:



Test Explorer view:



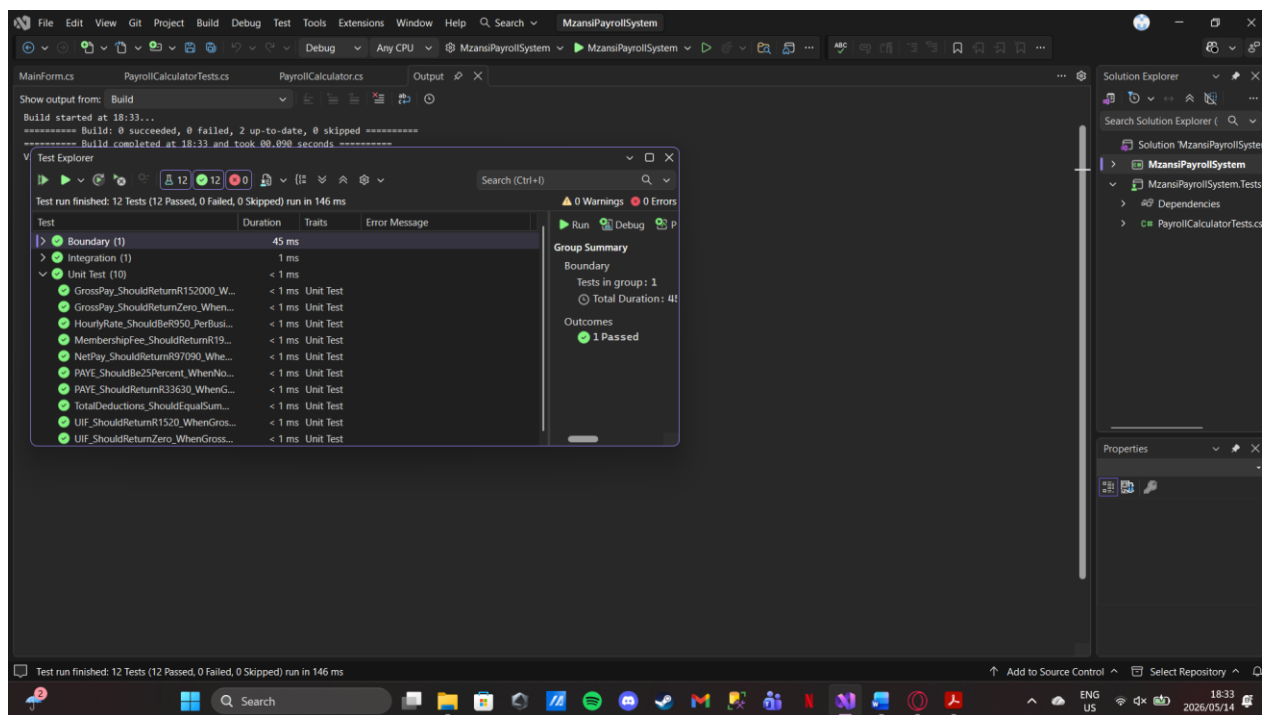
Fixed error:



The screenshot shows the Visual Studio IDE with the `PayrollCalculator.cs` file open. The code defines a `PayrollCalculator` class with several methods. A comment `//Fixed DEF-001` is visible next to the `CalculateUIF` method. The Solution Explorer on the right shows the project structure, including `MzansiPayrollSystem` and `PayrollCalculatorTests.cs`. The status bar at the bottom indicates "No issues found".

```
1 namespace MzansiPayrollSystem;
2
3
4
5 public class PayrollCalculator
6 {
7     // Hourly rate in South African Rand.
8     public const double HourlyRate = 950.00;
9
10    // Gross Pay = Hours Worked x R950.00
11    public double CalculateGrossPay(double hoursWorked) =>
12        hoursWorked * HourlyRate;
13
14    // UIF = 1% of Gross Pay
15    public double CalculateUIF(double grossPay) =>
16        grossPay * 0.01; //Fixed DEF-001
17
18    // Membership Fee = 13% of Gross Pay.
19    public double CalculateMembershipFee(double grossPay) =>
20        grossPay * 0.13;
21
22    public double CalculatePAYE(double grossPay, int numberOfDependents)
23    {
24        double taxableIncome = grossPay - (grossPay * 0.0575 * numberOfDependents);
25        return taxableIncome * 0.25;
26    }
27
28    //Total Deductions = UIF + PAYE + Membership Fee.
29    public double CalculateTotalDeductions(double uif, double paye, double membershipFee) =>
30        uif + paye + membershipFee;
```

Test explorer view

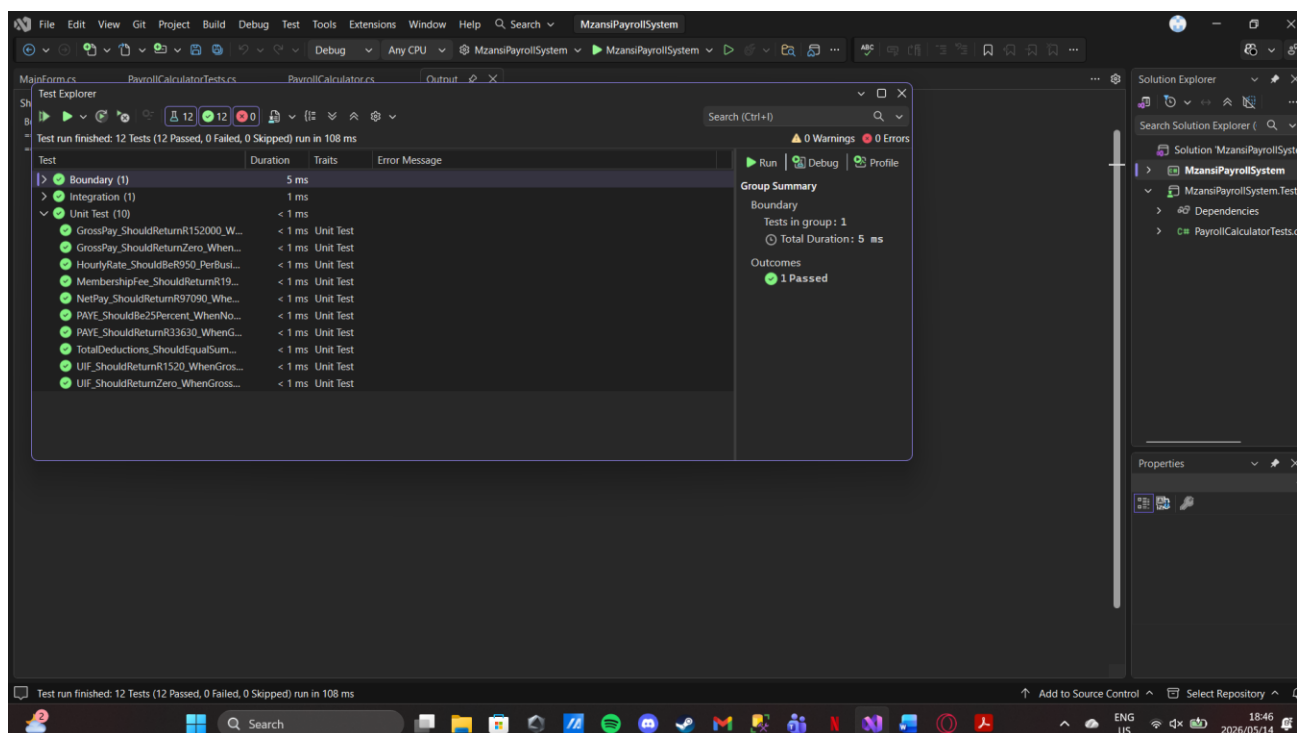


The screenshot shows the Visual Studio IDE with the Test Explorer view open. The test run finished with 12 tests passed, 0 failed, and 0 skipped. The test results are displayed in a table with columns for Test, Duration, Traits, and Error Message. The Group Summary on the right shows "1 Passed".

Test	Duration	Traits	Error Message
Boundary (1)	45 ms		
Integration (1)	1 ms		
Unit Test (10)	< 1 ms		
GrossPay.ShouldReturnR152000_W...	< 1 ms	Unit Test	
GrossPay.ShouldReturnZero_When...	< 1 ms	Unit Test	
HourlyRate.ShouldBeR950_PerBusi...	< 1 ms	Unit Test	
MembershipFee.ShouldReturnR19...	< 1 ms	Unit Test	
NetPay.ShouldReturnR97090_Wh...	< 1 ms	Unit Test	
PAYE.ShouldBe25Percent_WhenNo...	< 1 ms	Unit Test	
PAYE.ShouldReturnR33630_WhenG...	< 1 ms	Unit Test	
TotalDeductions.ShouldEqualSum...	< 1 ms	Unit Test	
UIF.ShouldReturnR1520_WhenGros...	< 1 ms	Unit Test	
UIF.ShouldReturnZero_WhenGros...	< 1 ms	Unit Test	

5.) Regression Testing

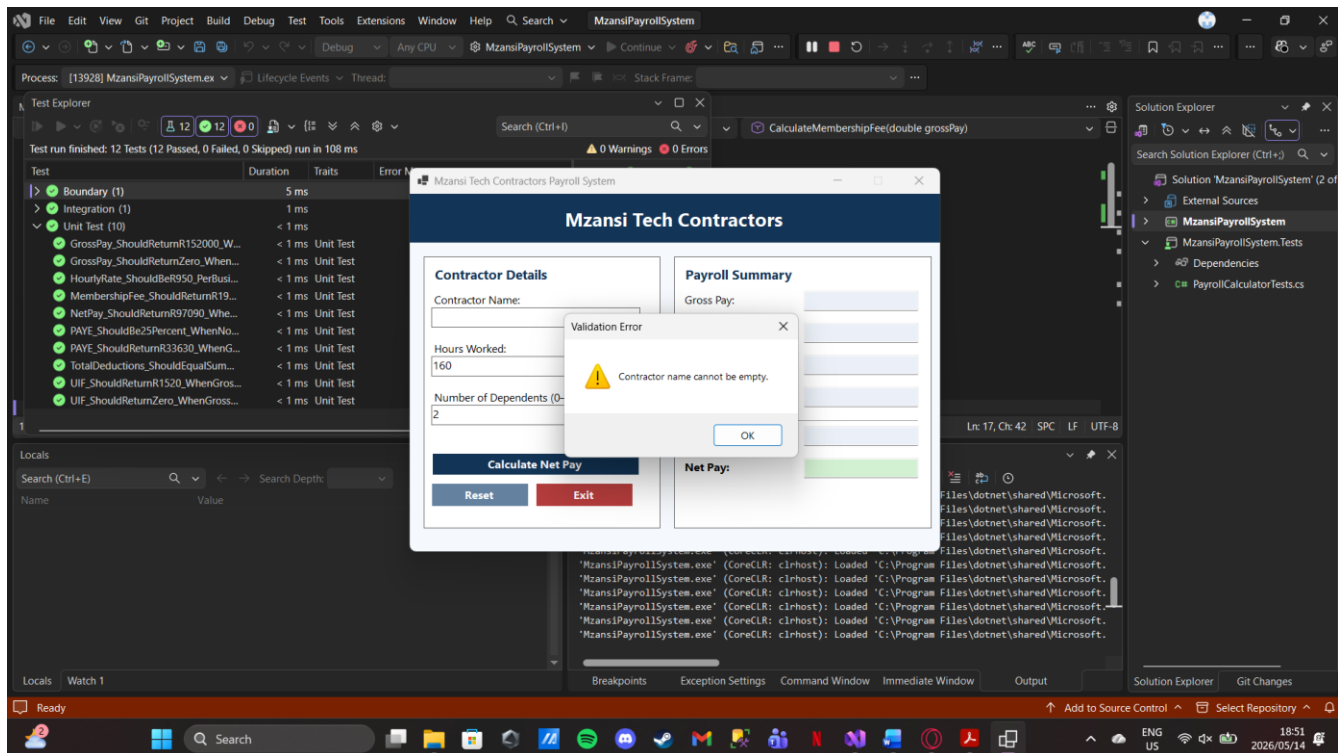
Test	Category	Result after fix
GrossPay_ShouldReturnR152000	Unit Test	PASS — unaffected
GrossPay_ShouldReturnZero	Unit Test	PASS — unaffected
UIF_ShouldReturnR1520	Unit Test	PASS — fixed
UIF_ShouldReturnZero	Unit Test	PASS — fixed
MembershipFee_ShouldReturnR19760	Unit Test	PASS — unaffected
PAYE_ShouldReturnR33630	Unit Test	PASS — unaffected
PAYE_ShouldBe25Percent	Unit Test	PASS — unaffected
NetPay_ShouldReturnR97090	Unit Test	PASS — fixed
FullPayrollFlow	Unit Test	PASS — fixed
PAYE_ShouldStayNonNegative	Unit Test	PASS — unaffected
HourlyRate_ShouldBeR950	Unit Test	PASS — unaffected



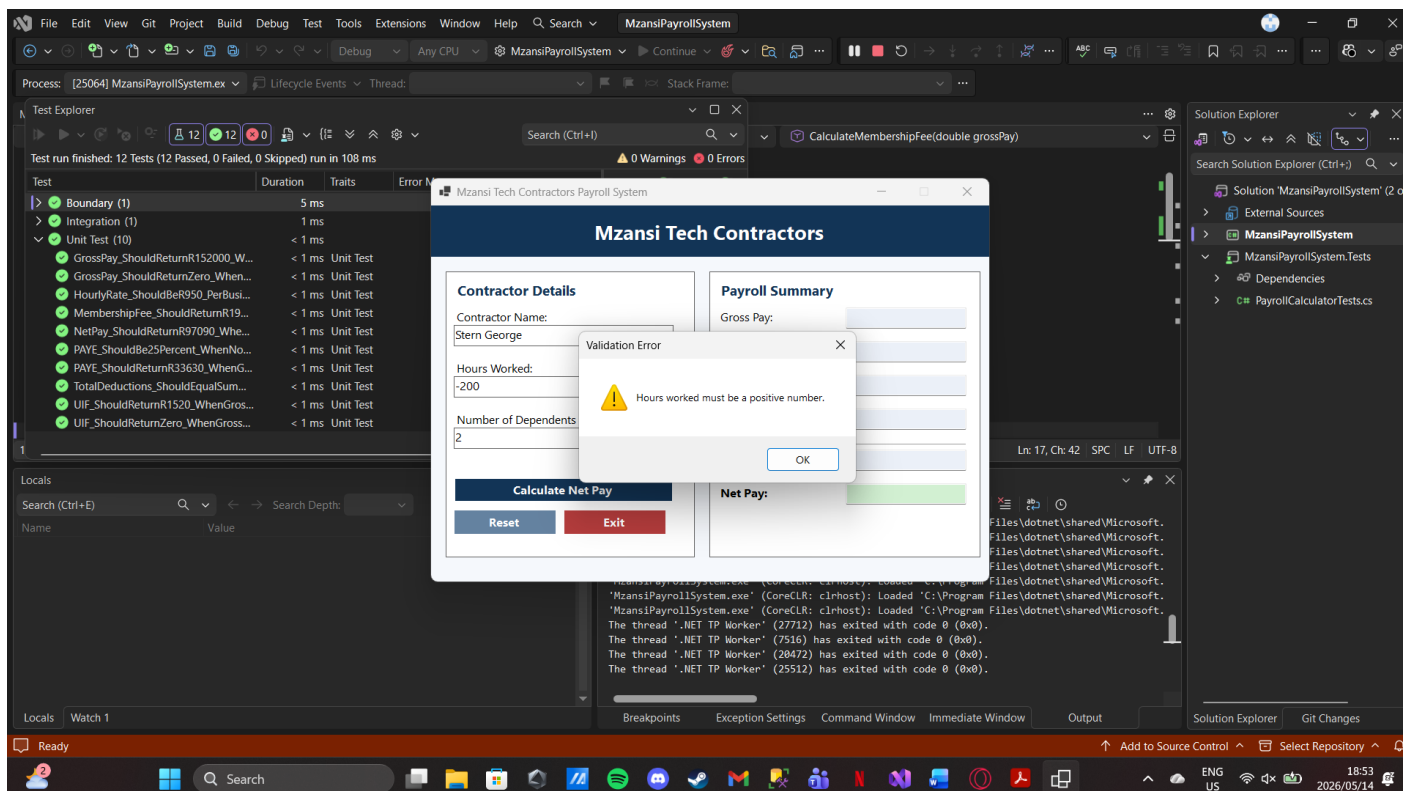
After the UIF correction, all 11 of the tests were executed to confirm no regression was introduced. The fix isolated the CalculateUIF method and had no negative impact on gross pay, PAYE, membership fee or net pay calculations.

Validation Checks:

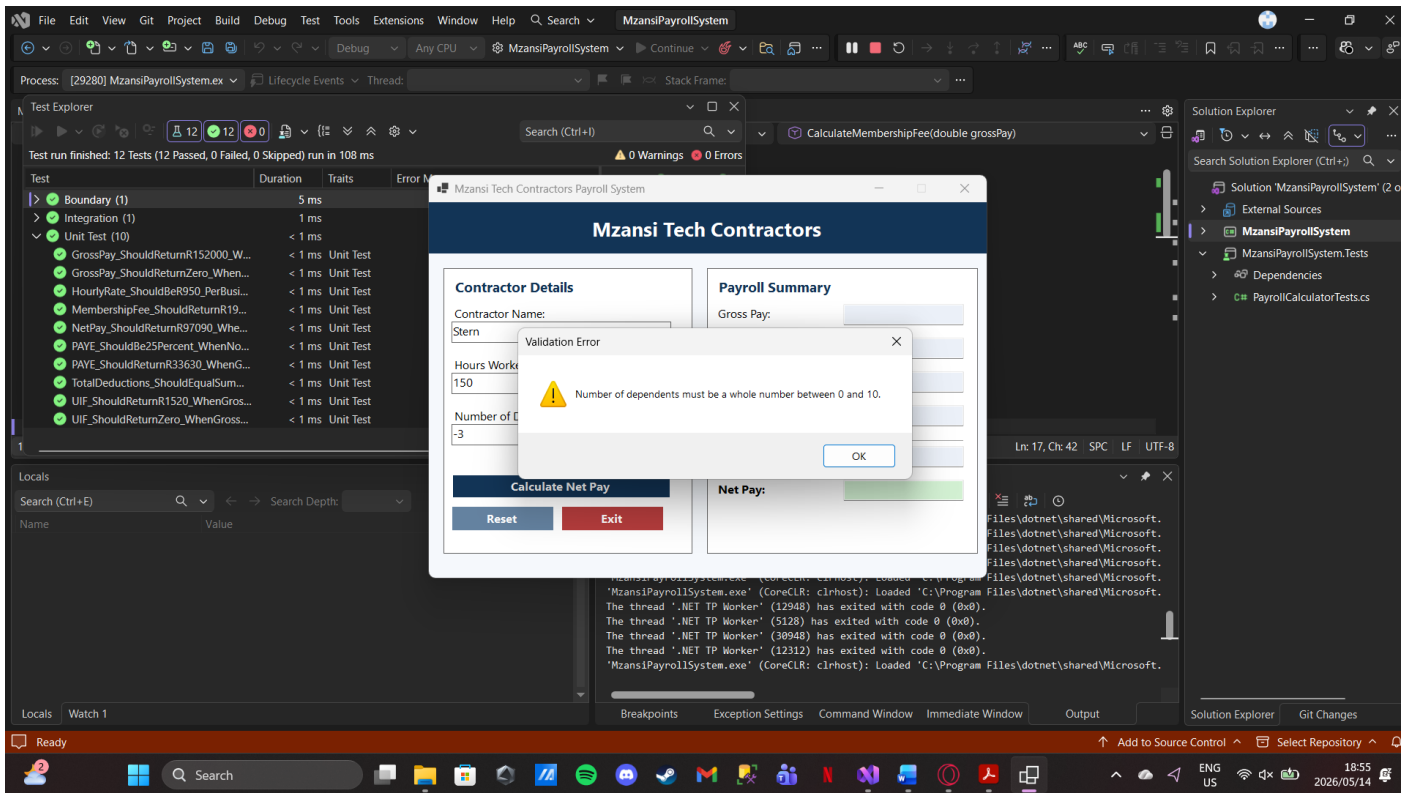
Constructor name can not be empty:



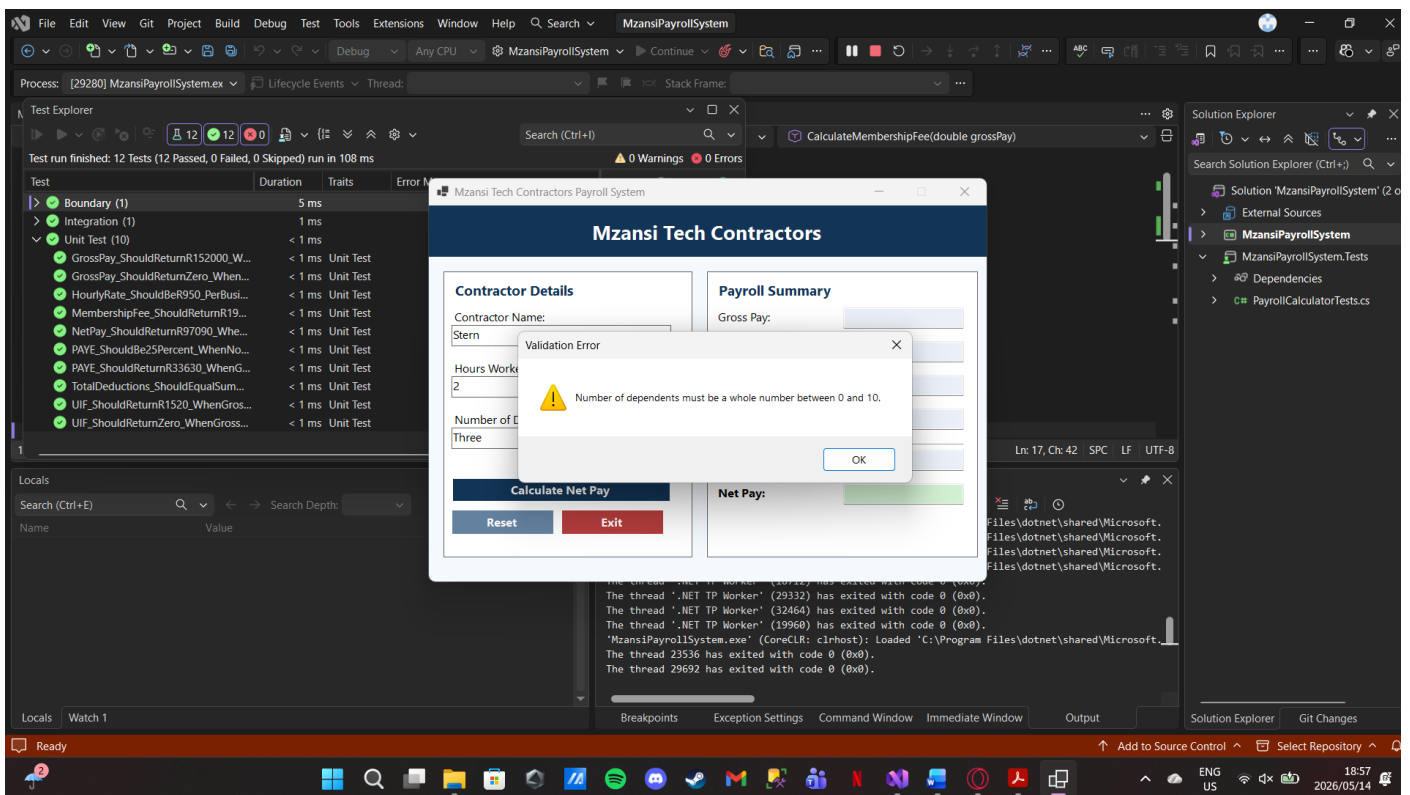
Hours worked can not be negative:

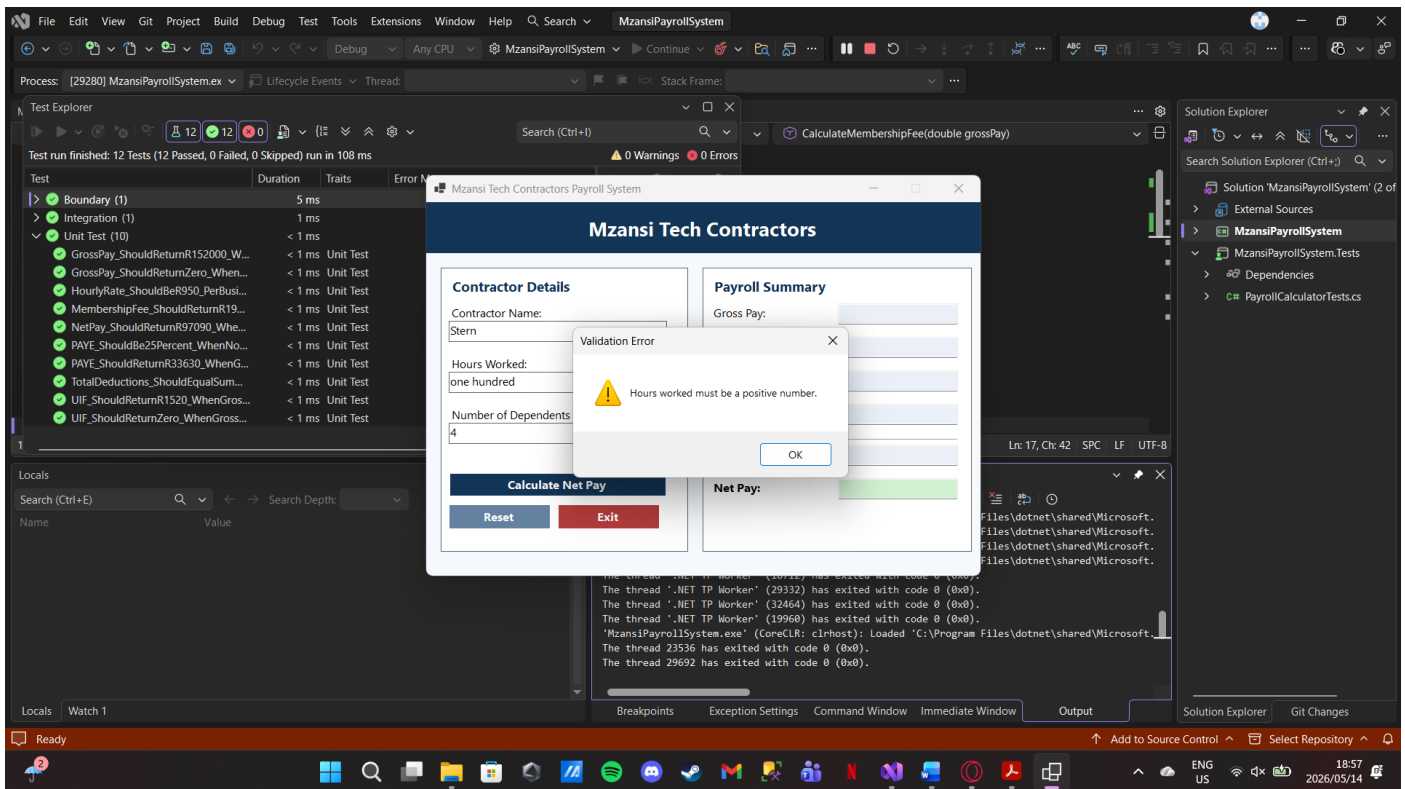


Number of dependants can not be negative:

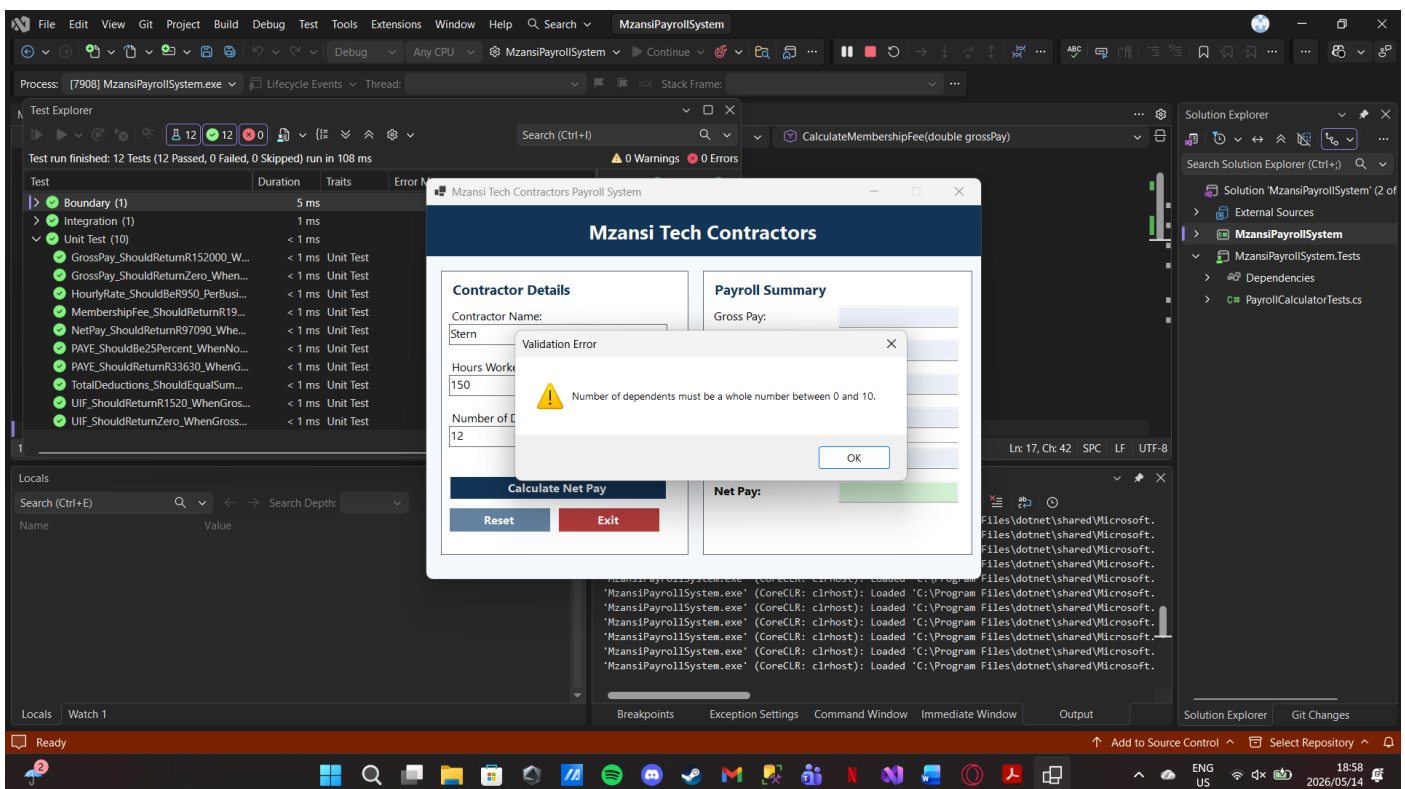


Non numeric values for hours or dependents





Dependents can not be greater than 10:



GitHub Link:

<https://github.com/SternGeorge17/PM04>